

Q1. Smart Document Editor using String, StringBuilder and StringBuffer

CODE

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a paragraph:");
        String paragraph = sc.nextLine();

        int characters = paragraph.length();

        String[] words = paragraph.trim().split("\\s+");
        int wordCount = paragraph.trim().isEmpty() ? 0 : words.length;

        String[] sentences = paragraph.split("[.!?]");
        int sentenceCount = 0;
        for (String s : sentences) {
            if (!s.trim().isEmpty())
                sentenceCount++;
        }

        System.out.println("Enter word to replace:");
        String oldWord = sc.next();

        System.out.println("Enter new word:");
        String newWord = sc.next();

        String replaced = paragraph.replace(oldWord, newWord);

        String[] arr = replaced.split("(?<=[.!?])");
        StringBuilder finalDoc = new StringBuilder();

        for (String s : arr) {
            String text = s.trim();
            String rev = new StringBuilder(text).reverse().toString();
            finalDoc.append(rev).append(" ");
        }

        StringBuffer history = new StringBuffer();
        history.append("Original: ").append(paragraph).append("\n");
        history.append("Replaced '").append(oldWord).append("' with '").append(newWord).append("'");
        history.append("Sentences Reversed\n");

        System.out.println("\nOriginal Document:");
        System.out.println(paragraph);

        System.out.println("\nCharacters: " + characters);
        System.out.println("Words: " + wordCount);
        System.out.println("Sentences: " + sentenceCount);

        System.out.println("\nEditing History:");
        System.out.println(history);

        System.out.println("Final Document:");
        System.out.println(finalDoc.toString().trim());

        sc.close();
    }
}
```

Report

INPUT

Java is easy. Java is platform independent
Java
Python

OUTPUT

(no output)

Q2. Hospital Token Management System using List Interface

CODE

```
import java.util.*;

public class PatientDetails {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        List<String> patients = new ArrayList<>();

        System.out.print("Enter number of patients: ");
        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {

            System.out.println("\nEnter Details of Patient " + (i + 1));

            System.out.print("Patient ID: ");
            String id = sc.nextLine();

            System.out.print("Patient Name: ");
            String name = sc.nextLine();

            System.out.print("Department: ");
            String department = sc.nextLine();

            patients.add(id + " " + name + " " + department);
        }

        System.out.println("\nRegistered Patients:");
        for (String p : patients) {
            System.out.println(p);
        }

        System.out.print("\nEnter Patient ID to Update: ");
        String updateId = sc.nextLine();

        for (int i = 0; i < patients.size(); i++) {

            if (patients.get(i).startsWith(updateId + " ")) {

                System.out.print("Enter New Name: ");
                String newName = sc.nextLine();

                System.out.print("Enter New Department: ");
                String newDept = sc.nextLine();

                patients.set(i, updateId + " " + newName + " " + newDept);
                break;
            }
        }

        System.out.println("\nAfter Updating:");
        for (String p : patients) {
            System.out.println(p);
        }

        System.out.print("\nEnter Patient ID to Remove: ");
        String removeId = sc.nextLine();

        Iterator<String> it = patients.iterator();
```

Report

```

        while (it.hasNext()) {
            String p = it.next();

            if (p.startsWith(removeId + " ")) {
                it.remove();
                break;
            }
        }

        System.out.println("\nAfter Removing:");
        for (String p : patients) {
            System.out.println(p);
        }

        System.out.print("\nEnter Department to Search: ");
        String dept = sc.nextLine();

        System.out.println("\nPatients in " + dept + ":");

        for (String p : patients) {
            if (p.endsWith(dept)) {
                System.out.println(p);
            }
        }

        if (!patients.isEmpty()) {
            System.out.println("\nFirst Registered Patient:");
            System.out.println(patients.get(0));

            System.out.println("\nLast Registered Patient:");
            System.out.println(patients.get(patients.size() - 1));
        }

        sc.close();
    }
}

```

INPUT

```

3
101
Rahul
Cardiology
102
Priya
Neurology
103
Arun
Orthopedics
102
Priya Sharma
Neurology
103
Cardiology

```

OUTPUT

(no output)